

# Credible on Paper? Evaluating the Effect of Green Banking Policy on Firms' Green Investment in Emerging Economies

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## Abstract

More than sixty emerging and developing economies have adopted green banking guidelines through the Sustainable Banking and Finance Network (SBFN). The policy's theory of change holds that supervisory guidelines lead banks to price environmental risk, so that credit flows to firms investing in cleaner production. We evaluate two links in that chain. At firm level, using two independent World Bank Enterprise Survey samples (28,042 firms in 41 economies; 89,797 firms in 162 economies) and four estimators, bank credit is associated with 13-percentage-point higher green-practice adoption, but the association is the same in adopting and non-adopting economies (0.076 versus 0.072 in a causal forest) and does not vary with regulatory quality. At country level, a staggered-adoption event study of 217 economies finds no effect on renewable-energy share, and a CO<sub>2</sub> association that appears to reflect which economies join. The guidelines show no detectable effect on the margin they target. For policy making, the results suggest that green finance programmes should be judged on firm and environmental outcomes rather than on guideline adoption alone, and that guidelines aimed at banks may need complements aimed at firms, such as technical assistance or credit

guarantees.

*Keywords:* green banking policy, policy evaluation, theory of change, access to finance, staggered difference-in-differences, causal forest

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## 1. Introduction

Since 2012, more than sixty central banks and financial regulators in emerging and transition economies have joined the IFC-hosted Sustainable Banking and Finance Network (SBFN), issuing guidelines that direct commercial banks to price environmental risk into lending and to channel credit towards cleaner production. The network monitors this effort through policy outputs: which guidelines each member has issued, and how far it has moved along the network’s own progression framework (IFC 2025). It does not monitor whether firms change what they do. Whether a firm that can obtain a bank loan is any more likely to install pollution-control equipment or measure its own emissions because such a guideline exists is the question an evaluation of the policy has to answer.

We treat SBFN adoption as a policy intervention and evaluate it against its own theory of change (Section 2.1). That theory runs from a supervisory guideline, through banks pricing environmental risk and reallocating credit, to financed firms investing more in cleaner production, and from there to better aggregate environmental outcomes. If the chain works, two things should be observable: the link between bank credit and firms’ green investment should be stronger where a guideline is in force, and adopting economies’ environmental indicators should improve after adoption. We test both.

Country-level studies find that green-finance policy works better where

22 governance is stronger (Chi and Yang 2023; Jia et al. 2026; Shi, Zhu, and  
23 Khan 2024; Ullah, Zubir, and Ariff 2024), and firm-level studies find that  
24 financing constraints shape green investment (Costa et al. 2024; De Haas et  
25 al. 2025; Ghosh and Dutta 2022). Neither literature has tested, on firm-level  
26 data and across a broad cross-section of adopting economies, whether green  
27 banking guidelines strengthen the credit-to-green channel they are designed  
28 to activate.

29 We use two independently assembled World Bank Enterprise Survey (WBES)  
30 samples: a 41-economy primary sample of 28,042 firms surveyed in 2018–  
31 2020 with the Bank’s full Green Economy module, and a 162-economy global  
32 sample of 89,797 firms surveyed in 2021–2026 that extends to high-income  
33 economies outside the SBFN’s remit. Because SBFN status is constant within  
34 a country in a single survey wave, we combine a pooled logit benchmark with  
35 a Bayesian hierarchical model, a causal forest, and, on the larger sample,  
36 country fixed effects. Across all four estimators the answer is the same, and it  
37 is precise: in the causal forest the credit effect is 0.076 in adopting economies  
38 and 0.072 in non-adopting ones. A Callaway–Sant’Anna staggered-adoption  
39 event study of 217 economies then tests the aggregate outcome directly.

40 The paper contributes to the evaluation of green finance policy in four  
41 ways. It evaluates SBFN-style adoption against firm-level outcomes rather  
42 than against the policy-stage indicators the network reports or the bank-level  
43 lending data that dominate the green-credit literature. It tests the theory of  
44 change at two points, with different data, units of analysis and identification  
45 strategies. It replicates the firm-level result on an independent sample with  
46 a different outcome and country fixed effects. And because the estimates

are precise rather than merely insignificant, it can report a genuine absence of effect on the targeted margin, the kind of null result that matters for programme design but is rarely published.

Section 2 describes the SBFN framework and its theory of change. Section 3 develops the hypotheses, Section 4 sets out data and methods, Section 5 evaluates the firm-level credit channel and Section 6 the aggregate outcome. Section 7 draws lessons for policy and evaluation, and Section 8 concludes.

## **2. Green banking policy and its theory of change**

The SBFN, founded in 2012 as the Sustainable Banking Network and hosted by the International Finance Corporation, is the main forum through which financial regulators and banking associations in emerging and developing economies coordinate national sustainable-finance policy. Its roughly seventy member institutions come almost entirely from Africa, Asia, Latin America, the Middle East and non-EU Europe (IFC 2025); none of the sixteen EU member states in our primary sample is a member.

Membership is the network’s coarsest measure of adoption. Its Multi-Tiered Progress Framework places each member on a progression from ‘Commitment’ or ‘Formulating’, through ‘Developing’, to ‘Advancing’ and ‘Consolidating’. Our sample spans that whole range: Mongolia and Morocco had reached ‘Advancing’ by their 2019 surveys, whereas Jordan and the Kyrgyz Republic were still at ‘Commitment’. Several economies, including Kazakhstan, Serbia, Ukraine and most of the Western Balkans, joined only after their survey was fielded. We code these as non-adopters, because what matters is the policy environment in force when a firm answered the survey.

71 *2.1. The policy’s theory of change and what this evaluation tests*

72 The case that SBFN adoption should change firm behaviour rests on a  
73 supervisory mechanism with five links:

- 74 1. *Adoption.* A banking regulator or banking association issues environmental-  
75 and-social risk management guidelines.
- 76 2. *Bank response.* Supervised banks face a compliance incentive to price  
77 environmental risk into lending and, in more advanced frameworks, to  
78 report a green-lending share.
- 79 3. *Credit allocation.* Credit flows more readily to firms undertaking cleaner  
80 investment and less readily to heavy polluters.
- 81 4. *Firm response.* Firms with bank credit invest more in green practices  
82 than they would without the guideline, so the credit-to-green link is  
83 stronger where the guideline is in force (H2), and stronger still where  
84 the supervisor can enforce it (H3).
- 85 5. *Aggregate outcome.* Adopting economies’ environmental indicators im-  
86 prove relative to non-adopters after adoption.

87 Evidence on China’s green credit guidelines documents links 2 to 4 at  
88 bank and firm level (Dai et al. 2025; Gao et al. 2022; Huang, Gao, and Jia  
89 2023; Li et al. 2023), and studies of Bangladesh and Vietnam point the same  
90 way (Khairunnessa, Vazquez-Brust, and Yakovleva 2021; Nguyen and Phan  
91 2025). Whether the chain holds across a heterogeneous set of SBFN adopters  
92 is untested. This paper tests link 4 at firm level (Section 5) and link 5 at  
93 country level (Section 6). It does not observe links 2 and 3: the Enterprise  
94 Surveys record whether a firm holds bank credit, not the terms banks offered

95 or the green share of their lending. A null at link 4 therefore shows that the  
96 chain does not deliver its intended firm-level effect, but cannot on its own  
97 say where it breaks.

### 98 **3. Hypotheses**

99 Green investments such as energy-management systems, on-site renew-  
100 able generation and waste-minimisation retrofits carry a large up-front cost  
101 against savings realised over several years, the profile of investment most  
102 sensitive to external finance. Firm-level evidence is consistent with this.  
103 Credit-constrained Indian manufacturers are less likely to adopt cleaner pro-  
104 duction (Ghosh and Dutta 2022); financial and managerial barriers jointly  
105 constrain green investment across countries (De Haas et al. 2025; Kalantzis,  
106 Schweiger, and Dominguez 2022); financing constraints blunt firms’ response  
107 to environmental policy in OECD and Vietnamese data (Costa et al. 2024;  
108 Truong 2026); and firms with better credit access sustain green management  
109 through shocks (Aristei and Gallo 2023). The WBES Green Economy mod-  
110 ule has been used to study such behaviour across many countries (Asif et al.  
111 2025; Phan 2026). The first-order proposition is:

112 **H1.** Firms with access to formal bank credit (a credit line or  
113 loan) are more likely to have adopted green production practices  
114 than otherwise similar firms without it.

115 H1 describes a channel, not a policy. The premise of SBFN-style guide-  
116 lines is that a supervisory mandate should widen the gap in green investment  
117 between firms with and without bank credit, because banks in adopting ju-  
118 risdictions have an incentive to seek out and price favourably exactly the

119 borrowers making such investments. Most supporting evidence comes from  
120 single-country studies of China’s unusually well-resourced regime (Dai et al.  
121 2025; Gao et al. 2022). Whether the same moderation holds across the  
122 far more heterogeneous, and often less advanced, set of SBFN adopters is  
123 untested:

124       **H2.** The association between bank credit and green-practice  
125 adoption (H1) is stronger in economies that had adopted an  
126 SBFN framework by the survey year than in economies that had  
127 not.

128 A guideline is only as effective as the supervisor behind it. The law-  
129 and-finance and state-capacity literatures show that formally identical rules  
130 produce different outcomes depending on the institutions that enforce them  
131 (Acemoglu, Johnson, and Robinson 2001; La Porta et al. 1998; Levine 2005).  
132 Where examination authority, sanctions and disclosure enforcement are weak,  
133 an SBFN framework may exist on paper without changing lending. Country-  
134 level evidence fits this view: green finance improves environmental outcomes  
135 only where corruption is controlled (Shi, Zhu, and Khan 2024), China’s guide-  
136 lines work best where environmental governance is strong (Chi and Yang  
137 2023), and governance strengthens the green-credit-to-sustainable-production  
138 link in BRICS economies (Jia et al. 2026). Each result implies an untested  
139 firm-level prediction. Proxying regulatory capacity by the Worldwide Gov-  
140 ernance Indicators (WGI) Regulatory Quality estimate, we hypothesise a  
141 three-way interaction:

142       **H3.** The moderation in H2 increases with regulatory quality:

143 SBFN adoption strengthens the credit-to-green channel most where  
144 the supervisor has the capacity to make the guideline credible.

### 145 3.1. *An auxiliary, exploratory hypothesis*

146 Financial constraints bind environmental investment most tightly in smaller  
147 firms (Ullah 2025), consistent with a fixed cost that is proportionately heavier  
148 for them even once financed. We did not pre-specify a firm-size hypothesis. It  
149 emerged from the causal forest (Section 5.4), and we report it as exploratory:

150 **H4 (exploratory).** Heterogeneity in the credit-to-green chan-  
151 nel is more attributable to firm size than to either country-level  
152 moderator.

## 153 4. Data and methods

### 154 4.1. *Data*

155 The WBES are stratified random-sample surveys of formal, non-agricultural  
156 private firms with at least five employees. Our primary sample is the Bank’s  
157 harmonised file for 41 Europe and Central Asia and Middle East and North  
158 Africa economies surveyed with an identical questionnaire in 2018–2020, the  
159 wave in which the Green Economy module was fielded in full: 28,042 firms  
160 (Supplementary Table S2). Six further economies (Bangladesh 2022, India  
161 2022, Indonesia 2023, Peru 2023, the Philippines 2023 and Timor-Leste 2021;  
162 15,556 firms) fielded a shortened, partly randomised version of the module.  
163 Because item wording differs, we do not pool them, and use them only for a  
164 narrower check (Section 5.5).

165 The global sample comes from the Bank’s standardised cross-country  
166 database, which harmonises core variables across 471 country-years and 292,314  
167 firms. In it, 160 country-years fielded in 2021–2026, including high-income  
168 economies such as Australia, Canada, Germany and Japan, carry CO2-  
169 emissions monitoring alongside the standard access-to-finance questions. Adding  
170 Bangladesh 2022 and Indonesia 2023 from our own extraction of the country  
171 files gives 162 economies and 100,115 firms, of which 89,797 have complete  
172 data. Controls come from the Bank’s Productivity Database, merged by firm  
173 identifier.

174 SBFN status, join year and progression stage for the primary sample  
175 are compiled from the network’s Global Progress Reports (2018, 2019, 2021,  
176 2023–2024), country addenda and data portal, cross-checked against IFC  
177 accession announcements. For the global sample we use the portal’s mem-  
178 ber roster (67 members). In both, an economy counts as an adopter only  
179 if it joined before its survey year. Six CEMAC economies and four East-  
180 ern Caribbean economies appear on the roster under regional entries, which  
181 we read as covering their common banking supervisors; these are inferences  
182 rather than per-country listings. Regulatory quality is the WGI estimate  
183 for the survey year (or the latest earlier year); WGI does not cover Taiwan,  
184 which is dropped.

185 A staggered difference-in-differences design around each economy’s join  
186 date is not feasible at firm level. The Green Economy module has been fielded  
187 to most economies only once, and economies surveyed twice by the core  
188 instrument did not repeat a comparable module, so no consistent outcome is  
189 observed before and after adoption. We therefore run the staggered design

190 at country level (Section 6) and treat the firm-level cross-section as a test of  
 191 link 4.

## 192 4.2. Variables

193 The outcome, *green practice adoption*, equals one if a firm adopted at  
 194 least one of seven practices in the previous three years: climate-friendly on-  
 195 site energy generation, an energy-management system, waste minimisation or  
 196 recycling, air-pollution control, other pollution control, any energy-efficiency  
 197 measure, or on-site renewable energy. The mean of the seven items serves as  
 198 a continuous index for robustness. *Bank credit* equals one if the firm holds a  
 199 credit line or loan from a formal financial institution. *SBFN member* equals  
 200 one if the economy had adopted an SBFN framework by the survey year, and  
 201 *regulatory quality* is the WGI estimate, standardised within the sample for  
 202 the hierarchical and machine-learning models. Controls are firm size (small,  
 203 medium or large), broad sector, log sales, exporter status and any foreign  
 204 ownership. Supplementary Table S1 gives the WBES code and source of  
 205 every variable.

## 206 4.3. Estimation

207 **Stage 1: pooled logit benchmark.** We estimate

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

208 for firm  $i$  in country  $c$ , with firm controls  $\mathbf{X}_{i,c}$ , sector fixed effects  $\delta_{s(i)}$   
 209 and country-clustered standard errors. A country fixed effect would absorb  
 210  $\text{SBFN}_c$ , so  $\beta_2$  would be unidentified. The interaction  $\beta_3$  would remain iden-  
 211 tified, because credit varies within countries, but only from the contrast be-

212 tween eight adopter economies and thirty-three non-adopters, and the level  
 213 difference between them would be lost. We therefore estimate  $\beta_2$  and  $\beta_3$   
 214 without country fixed effects, and add a triple interaction with regulatory  
 215 quality to test H3 directly (column M3).

216 **Stage 2: Bayesian hierarchical model.** Country enters as a random  
 217 effect, with a random slope on credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

218 estimated by the No-U-Turn Sampler in PyMC (Salvatier, Wiecki, and Fon-  
 219 nesbeck 2016). Partial pooling keeps the country-level main effects identified  
 220 while giving more defensible inference than cluster-robust standard errors  
 221 over only 41 clusters (Cameron and Miller 2015; Gelman and Hill 2007).  $\beta_3$   
 222 tests H2;  $\beta_4$  tests whether regulatory quality alone moderates the credit-  
 223 green link. Neither is a test of H3, which needs a three-way term that eight  
 224 adopter economies cannot support under partial pooling. H3 is tested by  
 225 Stage 1's column M3 and by the saturated global-sample model.

226 **Stage 3: causal forest.** To avoid imposing a linear interaction, we  
 227 estimate a causal forest with double machine learning (Athey, Tibshirani,  
 228 and Wager 2019; Chernozhukov et al. 2018; Wager and Athey 2018), using  
 229 gradient-boosted nuisance models and firm size, export status, SBFN status  
 230 and regulatory quality as effect modifiers. Its identifying assumption is the

231 same unconfoundedness on observables as Stages 1 and 2, so its conditional  
 232 average treatment effects (CATEs) are flexible conditional associations, not  
 233 effects identified from exogenous variation.

234 **Stage 4: global sample.** We re-estimate the model on the global sample  
 235 with CO2-emissions monitoring as the outcome. With 159 country clusters,  
 236 country fixed effects become usable: they absorb the SBFN and regulatory-  
 237 quality main effects but leave their interactions with credit identified. We  
 238 report this alongside a saturated specification without fixed effects that re-  
 239 tains the three-way term.

## 240 5. Evaluating the firm-level credit channel

### 241 5.1. Descriptive patterns

242 Firm counts in the primary sample range from 150 (Montenegro) to 3,075  
 243 (Egypt), and the unconditional adoption rate from 30.3% (Egypt) to 90.1%  
 244 (Malta). Eight of the 41 economies had adopted an SBFN framework by  
 245 their survey year: Egypt, Georgia, Jordan, the Kyrgyz Republic, Mongolia,  
 246 Morocco, Tunisia and Türkiye (Figure 1). Bank credit correlates with the  
 247 adoption index ( $r = 0.18$ ) but not with firms' perceived finance obstacle  
 248 ( $r = -0.02$ ). Figure 2 previews the main result. Plotting adoption against  
 249 regulatory quality separately for members and non-members gives two lines  
 250 of similar slope about 15–20 percentage points apart: a level difference, with  
 251 no difference in slope.

### 252 5.2. Pooled logit benchmark

253 Table 1 supports H1. Firms with a bank credit line are more likely to have  
 254 adopted a green practice ( $b = 0.571$ ,  $p < 0.001$ ), with an average marginal

255 effect of 12.95 percentage points (95% CI [7.7, 18.2]), about the same as  
 256 moving from a small to a large firm. Column M2 shows a large negative  
 257 SBFN main effect ( $b = -0.991$ ,  $p < 0.001$ ) and a small, insignificant credit  $\times$   
 258 SBFN interaction ( $b = 0.049$ ,  $p = 0.773$ ), contrary to H2. Column M3 tests  
 259 H3: the credit  $\times$  SBFN  $\times$  regulatory-quality term is small, of the wrong  
 260 sign and insignificant ( $b = -0.104$ ,  $se = 0.272$ ,  $p = 0.702$ ). Regulatory  
 261 quality’s own association with adoption is positive but imprecise ( $b = 0.278$ ,  
 262  $p = 0.110$ ).

### 263 5.3. Bayesian hierarchical model

264 The hierarchical model (Table 2; four chains, all  $\hat{R} \leq 1.01$ , bulk effective  
 265 sample sizes 797–2,270; Supplementary Section S7) gives H2 a fair hear-  
 266 ing without relying on 41-cluster standard errors. The credit effect survives  
 267 (posterior mean 0.33, 89% interval [0.18, 0.49]). SBFN membership lowers  
 268 baseline adoption ( $-0.57$ ,  $[-1.00, -0.09]$ ) and regulatory quality raises it  
 269 ( $0.34$ ,  $[0.16, 0.53]$ ), reproducing the level pattern in Figure 2. Neither inter-  
 270 action differs from zero: credit  $\times$  SBFN is  $0.072$  ( $[-0.26, 0.40]$ ) and credit  $\times$   
 271 regulatory quality is  $0.004$  ( $[-0.13, 0.14]$ ), a tight interval around zero rather  
 272 than merely a wide one.

### 273 5.4. Causal forest

274 The causal forest (Table 3, Figure 3) imposes no functional form on mod-  
 275 eration. Its average effect of credit is  $0.073$  (95% CI  $[-0.040, 0.186]$ ), smaller  
 276 and less precise than the logit marginal effect, as expected of a cross-fitted,  
 277 doubly robust estimator. The CATEs are flat across every institutional cut:

278 0.072 in non-adopting and 0.076 in adopting economies, and 0.075, 0.070 and  
279 0.075 across the low, middle and high regulatory-quality terciles.

280 Firm size accounts for 75.6% of the forest’s explained heterogeneity, against  
281 19.9% for regulatory quality, 2.6% for SBFN status and 1.9% for exporter  
282 status, consistent with H4. But the size strata do not show a monotonic gra-  
283 dient (0.080 for small, 0.075 for medium and 0.057 for large firms, all with  
284 overlapping intervals that include zero; Supplementary Table S3). We can  
285 say that heterogeneity sits at firm level rather than with country institutions,  
286 but not in which direction it runs.

## 287 5.5. *Robustness*

288 The result holds with the continuous adoption index estimated by OLS,  
289 and separately for manufacturing and services (Supplementary Table S5). In  
290 four of the six supplementary economies, an overdraft facility is associated  
291 with higher waste-minimisation adoption ( $b = 0.422$ ,  $p < 0.001$ ,  $N = 9,038$ ;  
292 Supplementary Table S4), so the credit channel is not specific to the primary  
293 region. One interaction is nominally significant: with overdraft access in  
294 place of a credit line, the overdraft  $\times$  SBFN term is negative ( $b = -0.346$ ,  $p =$   
295  $0.031$ ), the opposite of H2’s prediction. It is one of thirteen interaction tests  
296 under frequentist estimators, and after a Benjamini–Hochberg correction its  
297  $p$ -value rises to 0.403; no test in the family survives at a false discovery rate  
298 of 0.10 (Supplementary Table S6).

## 299 5.6. *Replication on the global sample*

300 The global sample differs from the primary one in outcome, period and  
301 economies, with little overlap (Figure 4). Without country fixed effects (Ta-

ble 4, M1), credit is associated with emissions monitoring ( $b = 0.198$ ,  $p = 0.039$ ; 89,797 firms, 159 countries), while the SBFN main effect ( $b = -0.171$ ,  $p = 0.353$ ), the credit  $\times$  SBFN interaction ( $b = 0.278$ ,  $p = 0.430$ ) and the regulatory-quality interactions are insignificant, including the three-way term this saturated model retains. With country fixed effects (M2), credit is more precisely estimated ( $b = 0.249$ ,  $p < 0.001$ ), and neither the credit  $\times$  SBFN ( $b = 0.506$ ,  $p = 0.205$ ) nor the credit  $\times$  regulatory-quality interaction ( $b = -0.044$ ,  $p = 0.346$ ) is significant. Figure 5 collects every estimate. The credit effect is positive in both samples and under every estimator, and every interaction interval includes zero.

## 6. Evaluating the aggregate outcome: a staggered-adoption event study

The firm-level evidence addresses link 4. Link 5, the aggregate outcome the policy is ultimately meant to move, can be tested directly using SBFN’s staggered join dates, because country-level outcomes are observed every year.

### 6.1. Data and design

We build a country-year panel of 217 economies for 2000–2024. Treatment timing is each economy’s SBFN join year from the portal roster, which agrees with our global-sample coding for all 61 overlapping economies. This gives 68 ever-treated economies (join years 2012–2024) and 149 never-treated ones. Outcomes from the World Development Indicators are renewable energy as a share of final energy consumption and CO2 emissions per capita; log GDP per capita (constant 2015 US dollars) is a covariate, because adopters are disproportionately fast-growing, industrialising economies.

326 We use the Callaway and Sant’Anna (2021) estimator, which estimates  
 327 group-time average treatment effects by adoption cohort and aggregates them  
 328 into an overall average treatment effect on the treated (ATT) and an event-  
 329 study profile, with never-treated economies as controls and 999 bootstrap  
 330 replications. Two-way fixed effects are biased under staggered adoption  
 331 (Goodman-Bacon 2021), so we report them only to show the size of that  
 332 bias. For each outcome we estimate an unconditional and a doubly robust  
 333 specification with log GDP per capita, and check the latter against a not-  
 334 yet-treated control group.

## 335 *6.2. Results*

336 Renewable-energy share shows no effect (Table 5):  $-1.192$  (95% CI  $[-2.763,$   
 337  $0.380]$ ) unconditionally and  $0.674$  ( $[-1.122, 2.470]$ ) with the GDP control,  
 338 and the event-study profile is flat on both sides of adoption (Figure 6). CO2  
 339 emissions per capita show a positive association, but three features argue  
 340 against a causal reading. It falls by about 45%, from  $0.631$  to  $0.349$ , once  
 341 GDP per capita is controlled for. Its event-study profile rises steadily from  
 342 near zero at adoption to a peak nine to twelve years later, which looks like  
 343 adopters already being on a steeper emissions path rather than like a policy  
 344 effect. And two-way fixed effects give  $0.901$ , more than double the doubly  
 345 robust estimate. Both doubly robust results are essentially unchanged with  
 346 not-yet-treated controls ( $0.313$  for CO2 and  $0.653$  for renewables). The design  
 347 finds no evidence that adoption improves an economy’s environmental tra-  
 348 jectory, and its one significant result is better explained by which economies  
 349 choose to join.

## 350 7. Discussion

351 In plain terms, a firm with a bank credit line is about 13 percentage points  
352 more likely to have adopted a green practice than a similar firm without one.  
353 This is an association, not a causal effect. The gap is the same whether  
354 or not the firm’s country has adopted an SBFN framework, and at every  
355 level of regulatory quality. Firms in adopting economies are instead 15–20  
356 percentage points less likely to have adopted any green practice at all: a  
357 difference in level, not in slope. Because the two firm-level samples share  
358 few economies, periods or outcome items, and the event study uses different  
359 data, units and identification, this convergence is unlikely to be an artefact  
360 of one sample or estimator.

361 The result complicates a green-credit literature built largely on China’s  
362 mature guidelines (Dai et al. 2025; Huang, Gao, and Jia 2023; Li et al.  
363 2023). Our adopters span every stage of SBFN’s progression framework, and  
364 across that range we find no sign that formal adoption changes how credit  
365 translates into green investment. ‘No interaction detected’ is not the same as  
366 ‘no complementarity exists’. A linear interaction may miss a threshold effect,  
367 given that SBFN’s framework is graded and our coding is binary, and the  
368 WGI may not capture banking-supervisory capacity. We consider our reading  
369 the best supported by the evidence, but it does not rule these alternatives  
370 out.

371 The one nominally significant interaction, a negative overdraft  $\times$  SBFN  
372 term, is at least consistent with guidelines steering credit towards term  
373 lending rather than towards working capital, although it does not survive  
374 multiple-testing correction. More generally, if smaller firms cannot absorb

the fixed costs of green investment even when financed, a guideline aimed at bank lending acts on the wrong margin however credible it is.

#### *7.1. Lessons for policy and for evaluation*

First, monitoring adoption is not monitoring effect. SBFN's progress framework scores members on the guidelines they issue and the stage they reach, and on our evidence being further along it does not coincide with any change in the firm-level margin the guidelines target. Programmes of this kind need outcome indicators alongside output indicators, at a minimum the share of bank credit financing green investment and firm-level adoption measured before and after a guideline takes effect.

Second, the data needed for a proper firm-level evaluation do not yet exist. Re-fielding a comparable Green Economy module in the same economies would allow the within-country before-and-after comparison this paper could run only at country level, and would let future evaluations locate where in the theory of change the chain breaks.

Third, bank-side guidelines may need firm-side complements, such as technical assistance, matching grants or credit guarantees for smaller firms, if the binding constraint lies with firms rather than with banks. The firm-size evidence here points that way but is exploratory.

## **8. Conclusion**

We evaluated SBFN green banking guidelines against their own theory of change. Bank credit is robustly associated with green-practice adoption, but that association is no stronger in adopting economies and does not depend on regulatory quality, on either of two independent firm-level samples and under

399 four estimators. A staggered-adoption event study finds no improvement in  
400 adopters' aggregate environmental outcomes. The policy implication is not  
401 that green banking guidelines are ineffective in every respect, but that our  
402 evidence gives no support to treating them as a lever that widens the green-  
403 investment gap between financed and unfinanced firms.

404 The evaluation has limits. The firm-level samples are cross-sections and  
405 cannot rule out reverse causality or unobserved country characteristics. SBFN  
406 membership is a binary proxy for a graded process, and WGI regulatory qual-  
407 ity is a broad governance measure rather than a supervisory one. The global  
408 sample's outcome is a single item. The design tests the intensive margin of  
409 the credit channel, not whether guidelines change who obtains credit, which  
410 our measure of realised access cannot observe, since some firms decline formal  
411 credit altogether (Jin 2026). And the event study, an aggregate test netted  
412 against one covariate, cannot exclude a small confounded relationship.

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414 The authors report there are no competing interests to declare.

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### 417 **Declaration of generative AI use**

418 The authors used generative AI tools to assist with drafting, editing and  
419 restructuring the manuscript text and with writing and checking analysis

code. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of the article.

## Data availability statement

The firm-level microdata are available from the World Bank Enterprise Surveys ([enterprisesurveys.org](https://enterprisesurveys.org); free registration required) under the Surveys' terms of use. The country-level datasets constructed for this paper and the full analysis code are available in a public repository; the link is withheld from this anonymised manuscript and will be provided to the editors on request. Supplementary material accompanies the article.

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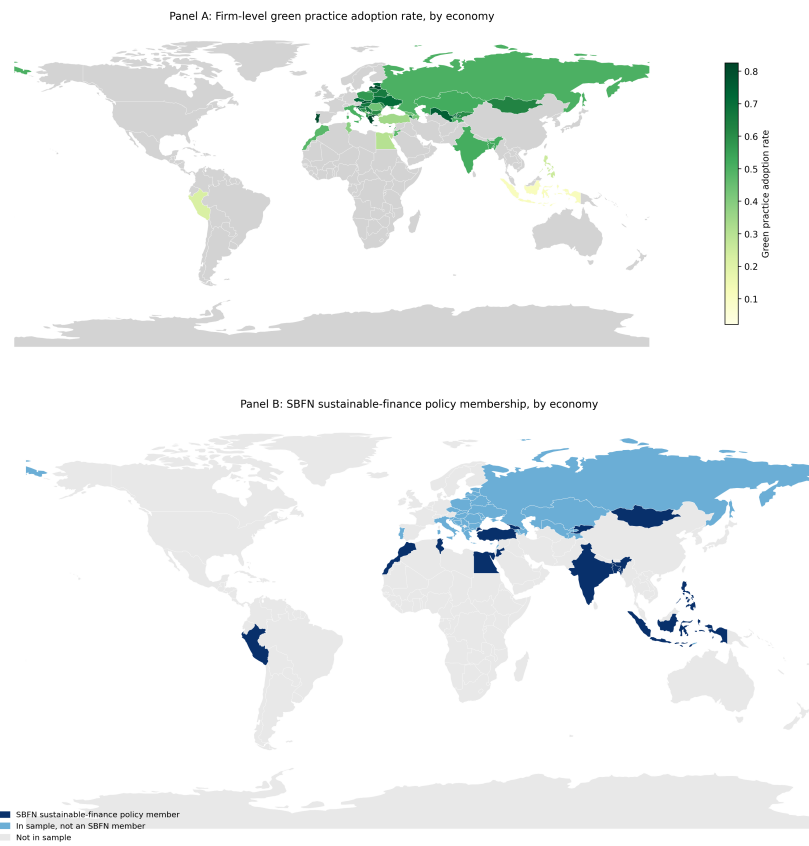


Figure 1: Green practice adoption and SBFN policy membership, by economy.

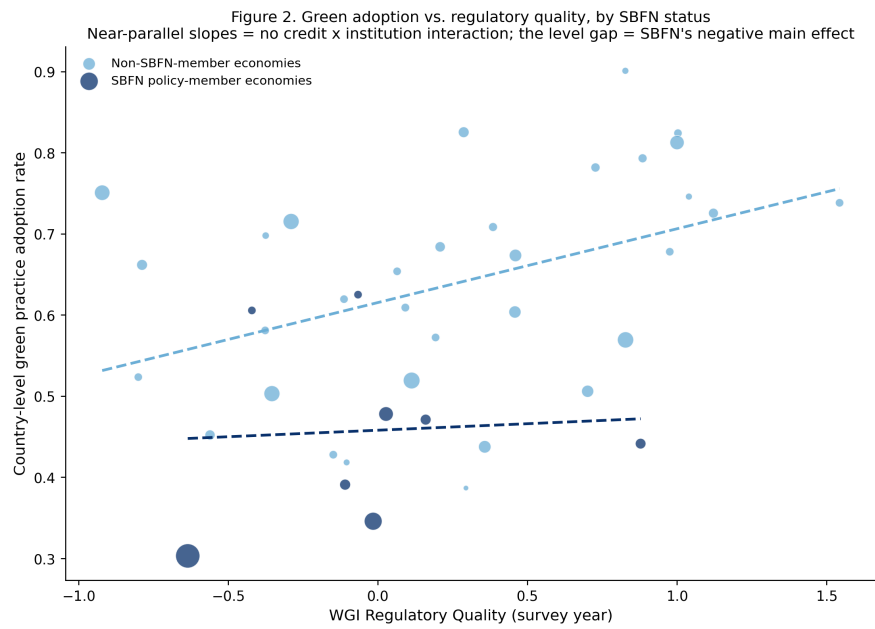


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

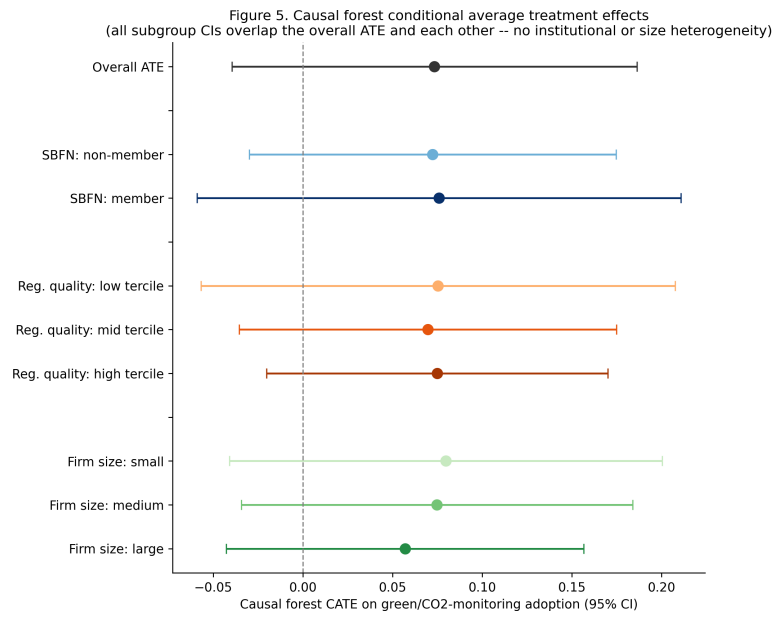


Figure 3: Causal forest conditional average treatment effects, by subgroup.

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

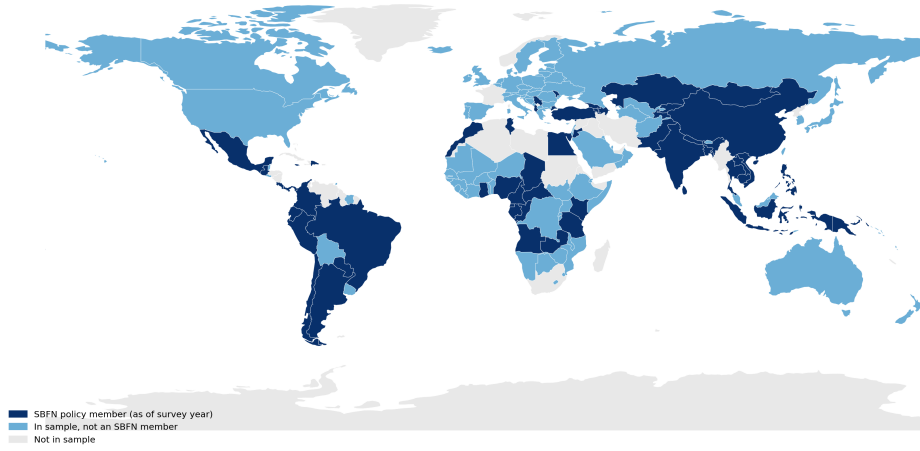


Figure 4: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

Figure 4: Coefficient estimates across every specification and both samples  
(Panel A: robust and significant everywhere; Panels B-C: null everywhere)

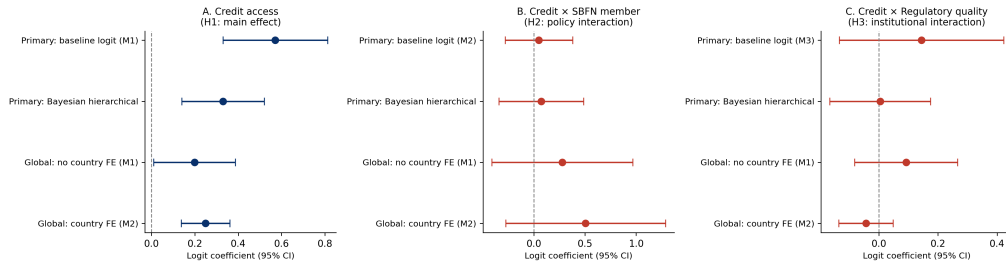


Figure 5: Coefficient estimates across every specification and both samples.

Country-level staggered event-study, SBFN adoption on macro environmental outcomes  
 (Callaway-Sant'Anna, doubly-robust, log GDP per capita control; shaded = pointwise 95% CI;  
 relative periods beyond  $\pm 12/9$  trimmed, too few surviving cohorts to be informative)

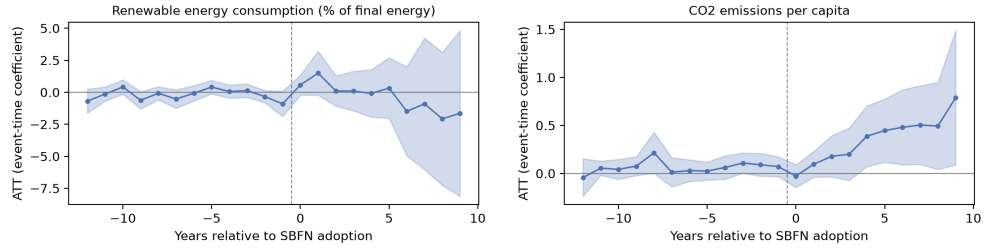


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond  $\pm 12/9$  are trimmed for having too few surviving cohorts to be informative.

Table 1: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

|                                            | M1                  | M2                   | M3                   |
|--------------------------------------------|---------------------|----------------------|----------------------|
| Credit access                              | 0.571***<br>(0.123) | 0.459***<br>(0.090)  | 0.373***<br>(0.085)  |
| SBFN member                                |                     | -0.991***<br>(0.178) | -0.844***<br>(0.209) |
| Credit $\times$ SBFN                       |                     | 0.049<br>(0.168)     | 0.019<br>(0.140)     |
| Regulatory quality                         |                     |                      | 0.278<br>(0.174)     |
| Credit $\times$ Reg. quality               |                     |                      | 0.144<br>(0.142)     |
| SBFN $\times$ Reg. quality                 |                     |                      | 0.053<br>(0.261)     |
| Credit $\times$ SBFN $\times$ Reg. quality |                     |                      | -0.104<br>(0.272)    |
| Log sales                                  | 0.061<br>(0.039)    | 0.041<br>(0.034)     | 0.068**<br>(0.032)   |
| Foreign owned                              | 0.219<br>(0.143)    | 0.244*<br>(0.139)    | 0.220<br>(0.143)     |
| Exporter                                   | 0.537***<br>(0.092) | 0.471***<br>(0.074)  | 0.412***<br>(0.079)  |
| Sector, size controls                      | Yes                 | Yes                  | Yes                  |
| $N$                                        | 23,914              | 23,914               | 23,914               |
| Pseudo $R^2$                               | 0.058               | 0.089                | 0.093                |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$

Table 2: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

| Parameter                    | Mean  | SD    | 89% ETI low | 89% ETI high | $\hat{R}$ |
|------------------------------|-------|-------|-------------|--------------|-----------|
| Intercept                    | -1.93 | 0.246 | -2.30       | -1.50        | 1.00      |
| Credit access                | 0.33  | 0.097 | 0.18        | 0.49         | 1.00      |
| SBFN member                  | -0.57 | 0.300 | -1.00       | -0.09        | 1.01      |
| Credit $\times$ SBFN         | 0.072 | 0.211 | -0.26       | 0.40         | 1.00      |
| Reg. quality (std.)          | 0.344 | 0.116 | 0.16        | 0.53         | 1.00      |
| Credit $\times$ Reg. quality | 0.004 | 0.087 | -0.13       | 0.14         | 1.00      |

Table 3: Causal forest DML: average treatment effect, CATEs, and feature importances.

| Estimate                          | Value      | 95% CI low | 95% CI high |
|-----------------------------------|------------|------------|-------------|
| Overall ATE                       | 0.0733     | -0.0396    | 0.1862      |
| CATE   SBFN non-member            | 0.0723     | -0.0300    | 0.1746      |
| CATE   SBFN member                | 0.0758     | -0.0591    | 0.2107      |
| CATE   reg. quality tercile: low  | 0.0753     | -0.0568    | 0.2075      |
| CATE   reg. quality tercile: mid  | 0.0697     | -0.0355    | 0.1749      |
| CATE   reg. quality tercile: high | 0.0749     | -0.0203    | 0.1701      |
| Feature                           | Importance |            |             |
| SBFN member                       | 0.0257     |            |             |
| Regulatory quality                | 0.1993     |            |             |
| Log sales                         | 0.7562     |            |             |
| Exporter dummy                    | 0.0189     |            |             |

Table 4: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

|                              | M1: no country FE   | M2: country FE      |
|------------------------------|---------------------|---------------------|
| Credit access                | 0.198**<br>(0.096)  | 0.249***<br>(0.057) |
| SBFN member                  | -0.171<br>(0.184)   | — (absorbed)        |
| Credit $\times$ SBFN         | 0.278<br>(0.352)    | 0.506<br>(0.399)    |
| Regulatory quality (std.)    | 0.155<br>(0.096)    | — (absorbed)        |
| Credit $\times$ Reg. quality | 0.092<br>(0.089)    | -0.044<br>(0.047)   |
| Log sales                    | 0.127***<br>(0.027) | 0.288***<br>(0.028) |
| Foreign owned                | 0.621***<br>(0.083) | 0.528***<br>(0.046) |
| Exporter                     | 0.456***<br>(0.094) | 0.335***<br>(0.044) |
| Country fixed effects        | No                  | Yes                 |
| <i>N</i>                     | 89,797              | 89,797              |
| Countries                    | 159                 | 159                 |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ . Taiwan dropped (no WGI coverage).

Table 5: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

| Outcome            | Specification                                                    | ATT       | SE    | 95% CI           |
|--------------------|------------------------------------------------------------------|-----------|-------|------------------|
| Renewable energy % | Callaway-Sant’Anna, no covariates, never-treated                 | -1.192    | 0.802 | [-2.763, 0.380]  |
| Renewable energy % | Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated   | 0.674     | 0.916 | [-1.122, 2.470]  |
| Renewable energy % | Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated | 0.653     | 0.919 | [-1.147, 2.454]  |
| Renewable energy % | Naive TWFE (post dummy, country+year FE)                         | -4.672*** | 1.127 | [-6.882, -2.463] |
| CO2 per capita     | Callaway-Sant’Anna, no covariates, never-treated                 | 0.631***  | 0.179 | [0.280, 0.982]   |
| CO2 per capita     | Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated   | 0.349**   | 0.156 | [0.044, 0.655]   |
| CO2 per capita     | Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated | 0.313**   | 0.148 | [0.023, 0.603]   |
| CO2 per capita     | Naive TWFE (post dummy, country+year FE)                         | 0.901***  | 0.204 | [0.500, 1.302]   |

\*\*\* 95% CI excludes zero; \*\* 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.